

SP Project Evaluations

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1 Q1

We consider a band-limited continuous-time signal $x(t)$ sampled at a rate above the Nyquist rate with sampling interval $T_s = 1/F_s$. The ideal samples are $x[n] = x(nT_s)$. Two types of non-ideal sampling are studied:

- (a) **Jittered sampling:** Each sample is taken with a timing error. The n -th sample is obtained as $\hat{x}[n] = x(nT_s + k_n\Delta)$, where $\Delta = T_s/10$ and the jitter values k_n are integers uniformly distributed in $[-K, K]$ for $1 \leq K \leq 4$.
- (b) **Missing samples:** With probability p a sample is unavailable. If $m[n] \in \{0, 1\}$ denotes the availability mask, then $\tilde{x}[n] = m[n]x[n]$, where $P(m[n] = 0) = p$ and $p \in (0.01, 0.10)$.

The objective in both cases is to reconstruct the ideal signal samples from the non-ideal observations.

1.1 Scenario (a): Reconstruction from Jittered Samples

Upsampling and Uniform Grid Representation: Due to jitter, the observed samples $\hat{x}[n] = x(nT_s + k_n\Delta)$ occur at non-uniform time instants. In order to apply Fourier-domain bandlimiting within the Papoulis–Gerchberg framework, we first construct a uniform high-rate time grid with spacing

$$T_{fine} = \frac{T_s}{10}.$$

This corresponds to an upsampling factor of 10 relative to the nominal sampling rate $1/T_s$.

The jittered samples are mapped onto this fine grid and an initial estimate $x^{(0)}(t_k)$ is obtained using PCHIP interpolation. All subsequent iterations of the reconstruction are performed on this upsampled uniform grid, while the jittered sample values are enforced at the nearest corresponding grid indices. This allows

us to convert the non-uniformly sampled problem into a uniform-domain band-limited reconstruction problem.

A fine uniform grid with spacing $T_{fine} = T_s/10$ is used to represent the signal. The jittered measurements $\hat{x}[n]$ are known along with their time instants. The reconstruction method follows the Papoulis–Gerchberg algorithm:

1. **Initialization:** Interpolate $\{\hat{x}[n]\}$ at the jittered time instants to obtain an initial estimate on the fine grid, using PCHIP interpolation.
2. **Bandlimit constraint:** Take the DFT of the estimate, apply an ideal low-pass mask corresponding to the signal bandwidth, and apply the inverse DFT.
3. **Data consistency constraint:** Replace the samples at jittered time instants with the known jittered observations.
4. **Iteration:** Steps 2–3 are repeated until convergence.

Reconstruction performance is evaluated using the mean absolute error (MAE) between the reconstructed and true fine-grid signals. The MAE is plotted as a function of jitter level K for three test signals.

1.2 Scenario (b): Reconstruction from Missing Samples

The signal is observed on the original sampling grid. A set of random samples are missing with probability p .

1. **Initialization:** Interpolate the available samples using PCHIP to fill missing points.
2. **Bandlimit constraint:** Apply FFT, enforce an ideal low-pass bandwidth B_W , and invert FFT.
3. **Data consistency:** Replace the values at available sample locations with the true samples, keeping the estimated values at missing locations.
4. **Iteration:** Repeat until convergence.

Performance is measured using MAE between the true and reconstructed discrete-time signals and plotted as a function of missing probability p .

1.3 Test Signals

Three different band-limited test signals composed of low-frequency sinusoids are used for demonstration. The results show increasing reconstruction error as the sampling becomes more non-ideal: MAE increases with larger jitter range K and with higher missing probability p .

2 Q2

- **Audio Input**

- Reads each audio file from a predefined list.
- Converts stereo signals to mono by averaging channels.

- **Time–Frequency Analysis (STFT)**

- Short-Time Fourier Transform (STFT) applied using a Hann window.
- Produces magnitude spectrogram $|S(f, t)|$, and time & frequency axes.

- **Onset Detection (Spectral Flux)**

- Computes frame-to-frame spectral changes:

$$SF(t) = \sum_f (S(f, t) - S(f, t - 1))^2$$

- Normalizes the novelty function to $[0, 1]$.
- Adaptive thresholding using mean + standard deviation:

$$Threshold = \mu + k\sigma$$

- Frames where spectral flux exceeds threshold are marked as drum hit onsets.

- **Hit Duration Estimation**

- For each onset, scan forward in time until spectral energy decays below 20% of peak.
- Duration computed as:

$$Duration = t_{end} - t_{onset}$$

- **Feature Extraction for Instrument Identification**

- Three spectral features are computed per hit:

1. **Spectral Centroid:** brightness of sound

$$C = \frac{\sum_f f \cdot S(f, t)}{\sum_f S(f, t)}$$

2. **Spectral Bandwidth:** spread of spectrum around centroid

$$B = \sqrt{\frac{\sum_f (f - C)^2 \cdot S(f, t)}{\sum_f S(f, t)}}$$

- 3. **Peak Frequency:** frequency bin with maximum magnitude
 - These features differentiate drums (kick = low frequency, snare = wideband, cymbal = high-frequency, etc.)
- **Unsupervised Clustering (Instrument Grouping)**
 - Hierarchical clustering using Euclidean feature distance:
 - * `pdist()` → compute pairwise feature distances
 - * `linkage()` → merge hits into clusters
 - * `cluster()` → assign cluster labels
 - Number of distinct instrument types = number of unique cluster labels.
- **Output and Visualization**
 - Prints hit index, hit time, duration, and assigned cluster group.
 - Scatter plot: Hit Time vs. Hit Duration, cluster-colored markers.
 - Allows visual validation of grouping (different instrument types exhibit distinct distributions).